Bangladesh Film Development Corporation (BFDC)

Portal API Integration & Dual-Direction Financial Data Exchange Standard

Feeder System for Accounting, General Ledger & Public Transparency

Document Version:	v1.1.0 (Draft Specification)	**Security Protocol:**	HTTPS / Bearer Token (Sanctum)
Integration Modes:	Dual (Scheduled Pull & Push)	**Data Serialization:**	Strict JSON (application/json)
Publish Date:	November 12, 2026	**Target Consumers:**	BFDC Finance & External Integration Partners

> **Executive Purpose:** This specification document defines the official interoperability contract between the **BFDC Portal** and external financial / accounting software (General Ledger, Billing & Collection ERP). It covers booking receipts, itemized bill breakdowns, NOC censor clearance payments, refund adjustments, master service categories, and both manual and automated synchronization mechanisms.

Overview: Summary of Available Endpoints (API Directory)

Below is the complete table of APIs exposed by the BFDC Portal for external consumers:

Method	Endpoint Path	Description & Operation	Direction
POST	`/api/login`	Acquires authenticated Bearer access token.	Inbound Auth
GET	`/api/item_categories`	Master List of 18 Service Categories & Departments.	Pull Master
GET	`/api/items` & `/api/items/{id}`	Master Catalogue of studio floors, cameras, lab services & rates.	Pull Master
GET	`/api/booking-payments`	List reconciled collections, bills & categorized line items (Feeder).	Pull Feed
GET	`/api/booking-payments/{trn_id}`	Single payment receipt & breakdown by transaction ID.	Pull Feed
GET	`/api/noc-payments`	List censor clearance & public NOC payment receipts.	Pull Feed
GET	`/api/noc-payments/{identifier}`	Single NOC transaction by token, TrxID, or NOC ID.	Pull Feed
GET	`/api/payment-refunds`	List cancelled bookings & net refund reversal adjustments.	Pull Adjust
GET	`/api/payment-refunds/{identifier}`	Single refund audit record by RFD code or transaction ID.	Pull Adjust
GET	`/api/parties` & `/api/parties/{id}`	Master ledger coordinates for producers & production banners.	Pull Master
POST	`/api/booking-payments/push`	Outbound trigger to dispatch booking payment to Accounting.	Push Outbound
POST	`/api/noc-payments/push`	Outbound trigger to dispatch NOC payment to Accounting.	Push Outbound
POST	`/api/payment-refunds/push`	Outbound trigger to dispatch refund credit note to Accounting.	Push Outbound

1. Master Service Categories (Departmental Chart of Accounts Mapping)

To eliminate mapping effort for external accounting ERPs, every bill line item carries both `category_id` and English `category`. Accounting systems can map these 18 master categories directly to their general ledger revenue codes / sub-ledgers:

Category ID	Category Name (Standard English)	Departmental / Accounting Scope	Items Count
1	**Spot & Floor Rental**	Studio Floors, Sound Stages & Spot Rentals	62 Items
2	**Camera & Primary Equipment**	Cinema Cameras & Primary Optical Units	20 Items
3	**Camera Accessories & Mounts**	Camera Lenses, Mounts, Rigs & Gimbal Accessories	46 Items
4	**Lighting Equipment**	Studio Lights, Generac Units & Electrical Kits	52 Items
5	**Set Construction Materials**	Set Building Materials, Flats & Structures	22 Items
6	**Video Editing & Post-Production**	Editing Suites, Offline & Online Suites	4 Items
7	**Sound Department**	Sound Recording, Dubbing & Foley Stage	10 Items
8	**Telecine Services**	Telecine Machine & Film Digital Scanners	6 Items
9	**Medical Equipment Hire**	On-set Medical Apparatus & Clinical Props	24 Items
10	**Floristry & Decor**	Flower Supplies & Decorative Elements	10 Items
11	**Laboratory & Color Grading**	Film Chemical Lab & Color Grading Suites	10 Items
12	**Subtitling & Translation**	Film Subtitling, Audio Transcription & Translation	4 Items
13	**Title Cards & Animation**	Opening Title Cards & 2D/3D Animation Lab	2 Items
14	**Technical Crew & Manpower**	Specialized Technical Manpower on Demand	16 Items
15	**Shooting Props & Dressing**	Shooting Props, Weapons, Furniture & Vehicles	128 Items
16	**Bangladesh Film City (Shooting)**	Kabirpur Film City Outdoor Shooting Sets	44 Items
17	**Film City (Events & Picnic)**	Film City Picnic Spots & Venue Hire	9 Items
18	**Composite Film Packages**	Comprehensive Package Deal Services	10 Items

> **Zero Manual Matching:** Instead of having to configure 500+ individual items manually, third-party software can simply map these **18 Categories** to their respective Revenue Heads. Any item under Category ID 1 automatically credits "Studio Floor Revenue", Category ID 2 credits "Camera Rental Revenue", and so forth.

2. Master Service Items Directory (`item_id` Mapping Guide)

When pulling booking line items (under `lineItems[].item_id`), each service is identified by a standardized item ID. Third-party accounting software can also retrieve the live list programmatically via `GET /api/items`:

ID	Category	BFDC Official Service / Item Name	Department	Unit Type
1	Cat 1	Floor No. 1 (2400 sq ft) Set Construction Rent (incl. 1 setman)	Studio Floor	Shift
2	Cat 1	Floor No. 1 Shooting Period Rent (1 makeup room, 30KW light, technicians)	Studio Floor	Shift
3	Cat 1	Floor No. 2 (without AC) (11750 sq ft) Set Construction Rent	Studio Floor	Shift
4	Cat 1	Floor No. 2 Shooting Period Rent (without AC)	Studio Floor	Shift
5	Cat 1	Floor No. 2 (with AC) (11750 sq ft) Set Construction Rent	Studio Floor	Shift
6	Cat 1	Floor No. 2 Shooting Period Rent (with AC)	Studio Floor	Shift
7	Cat 1	Floor No. 2 Wall Rent (including makeup room)	Studio Floor	Shift
8	Cat 1	Floor No. 6 (2400 sq ft) Set Construction Rent	Studio Floor	Shift
9	Cat 1	Floor No. 6 Shooting Period Rent	Studio Floor	Shift
10	Cat 1	Floor No. 7 (without AC) (3040 sq ft) Set Construction Rent	Studio Floor	Shift
11	Cat 1	Floor No. 7 Shooting Period Rent (without AC)	Studio Floor	Shift
12	Cat 1	Floor No. 7 (with AC) (3040 sq ft) Set Construction Rent	Studio Floor	Shift
13	Cat 1	Floor No. 7 Shooting Period Rent (with AC)	Studio Floor	Shift
14	Cat 1	Floor No. 8 (9280 sq ft) Set Construction Rent	Studio Floor	Shift
15	Cat 1	Floor No. 8 Shooting Period Rent	Studio Floor	Shift
75	Cat 2	4K Cinema Camera Rig Unit	Camera Department	Shift
120	Cat 4	Skypanel LED Softlight 300W	Lighting Department	Shift
210	Cat 7	Dolby Atmos Dubbing & Foley Stage	Sound Department	Hour

3. Protocol, Security & Connection Standards

Communication with the BFDC Portal occurs over standard HTTP REST interfaces using JSON payloads. All non-public endpoints require bearer authentication.

Parameter	Specification	Compliance Rule
Base URL (Staging)	`http://fdc.test/api`	Dedicated sandboxed instance for integration testing.
Base URL (Production)	`https://portal.bfdc.gov.bd/api`	Strictly TLS 1.3 / HTTPS encryption mandatory.

HTTP Headers	`Authorization: Bearer <token>`
	`Accept: application/json`
	`Content-Type: application/json`
		Required for all data-exchange endpoints. Missing tokens return `401 Unauthorized`.

Identifiers Standard	Enterprise Prefix Codes (e.g. `PRTY-00018`, `APP-00042`, `RFD-00004`)	Preserves database normalization while completely eliminating internal schema exposure.

3.1 Authentication Endpoint

`POST` `/api/login`

Integration clients submit authorized credentials to obtain a Bearer access token.

Request Payload:

```
{
        "email": "accounting@bfdc.gov.bd",
        "password": "IssuedPassword#2026"
}
```

Success Response (HTTP 200 OK):

```
{
        "success": true,
        "data": {
                "token": "1|eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
                "token_type": "Bearer",
                "name": "BFDC Accounting Integration Consumer"
        },
        "message": "User authenticated successfully"
}
```

4. Booking & Package Payments API (Feeder for Accounting)

This endpoint serves completed and verified receipts. In a single payload, it encapsulates the **Collection / Money Receipt**, the associated **Party / Production Banner**, and the itemized **Bill Memo (Line Items with Category ID)**.

GET /api/booking-payments | GET /api/booking-payments/{trn_id}

Query Parameters: `status=success|refund`, `date=YYYY-MM-DD`, `limit=20`

JSON Field	Type	Accounting Feeder Role	Sample / Description
`partyId`	String	General Ledger Party Code	`PRTY-00018` (Mapped to Master Producer Ledger).
`applicationId`	String	Project / Film Code	`APP-00042` (Universal Application reference).
`mrNo`	String	Money Receipt / Bank Settled No	`TRX-2026-9874561` (Reconciles bank statements).
`collectionAmount`	Number	Total Debit / Credit Value	`45000.00` (Expressed in BDT).
`jobReportNo`	String	Requisition Reference	`BKG-20260215-01` (Cross-references shift bookings).
`lineItems[]`	Array	Itemized Bill Memo	Breakdown with `category_id`, `category`, `subSection`, `rate`, `quantity`, `unit`.

Sample Response Payload (HTTP 200 OK):


```
{
        "success": true,
        "message": "Booking Payments retrieved successfully",
        "data": {
                "partyId": "PRTY-00018",
                "applicationId": "APP-00042",
                "mrNo": "TRX-2026-9874561",
                "collectionDate": "2026-11-11",
                "collectionAmount": 45000,
                "status": "success",
                "banner": "Impress Telefilm Limited",
                "filmName": "Nodir Desh",
                "producer": "Faridur Reza Sagar",
                "jobReportNo": "BKG-20260215-01",
                "billDate": "2026-11-11",
                "lineItems": [
                        {
                                "item_id": 12,
                                "category_id": 1,
                                "category": "Spot & Floor Rental",
                                "section": "Studio Floors",
                                "subSection": "Floor No. 7 (with AC) Set construction rent",
                                "shift": "Morning",
                                "rateType": "GeneralRate",
                                "rate": 15000,
                                "quantity": 3,
                                "unit": "Shift"
                        },
                        {
                                "item_id": 75,
                                "category_id": 2,
                                "category": "Camera & Equipment",
                                "section": "Camera Department",
                                "subSection": "4K Cinema Camera Rig Unit",
                                "shift": "Morning",
                                "rateType": "GeneralRate",
```

```
                                "rate": 20000,
                                "quantity": 1,
                                "unit": "Shift"
                        }
                ],
                "auditTrail": {
                        "trn_id": "EP_1794225024",
                        "TrxID": "TRX-2026-9874561",
                        "party_code": "PRTY-00018",
                        "application_code": "APP-00042",
                        "type": "booking",
                        "paidAt": "2026-11-11 12:30:15"
                }
        }
}
```

5. NOC (No Objection Certificate) Payments API

Provides financial tracking and revenue auditing for No Objection Certificate applications and public clearance services.

GET /api/noc-payments | **GET /api/noc-payments/{identifier}**

Query Parameters: `status=paid|pending|approved`, `date=YYYY-MM-DD`, `search=<text>`

JSON Field	Type	Role & Description
`nocId`	String	Unique NOC Universal Code (e.g. `NOC-00015`).
`mrNo`	String	Gateway Settlement Transaction ID (e.g. `TRX-789456123`).
`collectionAmount`	Number	Fee amount in BDT (e.g. `1000.00`).
`applicant`	Object	Includes applicant name, mobile number, organization, and address.
`nocDetails`	Object	Work/Film name, type, and official censor certificate number.

Sample Response Payload (HTTP 200 OK):


```
{
        "success": true,
        "message": "NOC Payments retrieved successfully",
        "data": {
                "nocId": "NOC-00015",
                "mrNo": "TRX-789456123",
                "collectionDate": "2026-11-11",
                "collectionAmount": 1000,
                "status": "paid",
                "serviceName": "NOC Application Fee",
                "applicant": {
                        "fullName": "Rahim Khan",
                        "designation": "Director",
                        "producer": "Impress Telefilm",
                        "organization": "Impress Group",
                        "mobile": "01755666666",
                        "email": "applicant@example.com",
                        "address": "Dhaka, Bangladesh"
                },
                "nocDetails": {
                        "filmOrWorkName": "Dui Prithibi",
                        "type": "Cinema",
                        "censorCertificateNo": "CEN-2026-001",
                        "censorCertificateDate": "2026-10-15",
                        "publishDate": "2026-11-01",
                        "place": "Dhaka"
                },
                "auditTrail": {
                        "noc_code": "NOC-00015",
                        "token": "NOC-TOKEN-123456",
                        "trx_id": "TRX-789456123",
                        "paidAt": "2026-11-11 14:30:00"
                }
        }
}
```

6. Payment Refunds & Cancellations API

Audits canceled bookings and custom package requisitions. Facilitates general ledger reversal by providing the gross paid amount, the 10% BFDC retained service charge, and the 90% net refunded amount.

GET /api/payment-refunds | **GET /api/payment-refunds/{identifier}**

Query Parameters: `status=pending|refunded`, `type=booking|package`, `date=YYYY-MM-DD`

Sample Response Payload (HTTP 200 OK):


```
{
        "success": true,
        "message": "Payment Refund retrieved successfully",
        "data": {
                "refundId": "RFD-00004",
                "originalTrnId": "TRN-17865268071562",
                "mrNo": "TRX-2026-123456",
                "partyId": "PRTY-00018",
                "applicationId": "APP-00042",
                "banner": "Impress Telefilm Limited",
                "filmName": "Nodir Desh",
                "producer": "Faridur Reza Sagar",
                "paymentType": "booking",
                "grossAmount": 10000,
                "deductedCharge": 1000,
                "netRefundAmount": 9000,
                "status": "refunded",
                "refundDate": "2026-11-11",
                "auditTrail": {
                        "refund_ref": "RFD-00004",
                        "trn_id": "TRN-17865268071562",
                        "payment_id": 42,
                        "requestedAt": "2026-11-11 15:10:00"
                }
        }
}
```


7. Party Master & Producer Directory API

Provides financial software with authenticated producer profile coordinates (Tax ID, Trade License, and Banking coordinates).

GET /api/parties | **GET /api/parties/{id}**

Sample Response Payload (HTTP 200 OK):


```
{
        "success": true,
        "data": {
                "partyCode": "PRTY-00018",
                "organization": "Impress Telefilm Limited",
                "ownerName": "Faridur Reza Sagar",
                "tinNumber": "458963214578",
                "tradeLicense": "TRAD/DNCC/021548/2026",
                "bankName": "Sonali Bank PLC",
                "accountNumber": "0124589632014",
                "phone": "01711000000",
                "status": "active"
        }
}
```

8. Dual-Direction Integration: Pull vs. Push Architecture

The BFDC Portal operates smoothly under both synchronization paradigms:

Mode A: Pull Integration (External Consumer Automated Ingestion)
The consumer system configures a scheduled chron-job / polling worker to query:

- `GET /api/booking-payments?status=success` → Ingests collections, line items, and memo bills.
- `GET /api/noc-payments?status=paid` → Ingests NOC revenue entries.
- `GET /api/payment-refunds?status=refunded` → Ingests reversal credit notes.

Mode B: Push Integration (BFDC Event-Driven & Administrative Triggers)

1. **Automatic Background Push (Real-Time):** Configurable via `BFDC_ACCOUNTING_AUTO_SYNC=true`. Whenever a citizen or producer completes payment or cancellation, the BFDC backend issues real-time HTTP requests to the target Accounting API.
2. **Administrative Web Push (Human-in-the-Loop):** If network outages occur, BFDC authorized officers can re-trigger on-demand push events via Web UI buttons:
   - **Payment List (`/makePayments`):** Click Action → **"Push to Accounting"** or **"Push Refund"**.
   - **NOC Paid List (`/nocApplication/paid`):** Click Action → **"Push to Accounting"**.
3. **Programmatic Push Endpoints:**
   - `POST /api/booking-payments/push` (Body: `{"trn_id": "TRN-..."}`)
   - `POST /api/noc-payments/push` (Body: `{"identifier": "NOC-..."}`)
   - `POST /api/payment-refunds/push` (Body: `{"identifier": "RFD-..."}`)

9. Audit, Governance & Cross-Reference Table

Reconciliation Entity	JSON Key	Audit Role & Verification Mechanism
Bank Settlement Number	`mrNo` / `TrxID`	Verifies money deposit against central payment gateway settlement batches and bank statements.
BFDC Invoice Tracking	`jobReportNo` / `trn_id`	Reconciles equipment hire, shooting floor usage, and lab services against internal booking sheets.
Party / Customer Ledger	`partyId` (e.g. `PRTY-00018`)	Maintains clean, deduplicated general ledger accounts across multiple film projects.
Refund Audit Reference	`refundId` (e.g. `RFD-00004`)	Ties credit note reversals to the exact initial transaction and tracks 10% retained administrative fee.